University of Central Punjab

Database System

JOIN Queries

Task 1

Write a query to display the name and transaction id of the employees whose transaction amount was greater than 100000

```
mysql> SELECT
    ->     e.name, t.transaction_id
    -> FROM
    ->     employee e
    -> INNER JOIN
    ->     transaction t ON e.eid = t.eid
    -> WHERE
    ->     t.transaction_amount > 100000;
+--------+----------------+
| name   | transaction_id |
+--------+----------------+
| Ayesha | tr102          |
| Anwar  | tr201          |
| Anwar  | tr206          |
| Kamran | tr205          |
+--------+----------------+
4 rows in set (0.00 sec)
```

Task 2

Write a query to display all the customer names with their saving accounts number having balance more than 10000and less than 50000.

```
mysql> SELECT c.name, s.Account_number
    -> FROM customer c
    -> JOIN savingaccounts s ON c.customer_id = s.customer_id
    -> WHERE s.current_balance > 10000
    ->   AND s.current_balance < 50000;
+-------------+----------------+
| name        | Account_number |
+-------------+----------------+
| Ali Ahmad   | sbABL107       |
| M. ALi      | sbABL303       |
| Shafqat Ali | sbMCB100       |
| Imtiaz Khan | sbMCB101       |
| M. Kareem   | sbUBL103       |
| Waqar Azeem | sbUBL105       |
+-------------+----------------+
6 rows in set (0.00 sec)
```

Task 3

Write a query to display all the saving account number with their owner's name.

```
mysql> SELECT
    ->     s.Account_number, c.name
    -> FROM
    ->     savingaccounts s
    -> INNER JOIN
    ->     customer c
    -> ON
    -> s.customer_id = c.customer_id;
+----------------+-------------+
| Account_number | name        |
+----------------+-------------+
| sbUBL204       | Imran Khan  |
| sbUBL103       | M. Kareem   |
| sbABL107       | Ali Ahmad   |
| sbUBL105       | Waqar Azeem |
| sbABL109       | Ahsan ALi   |
| sbABL303       | M. ALi      |
| sbMCB101       | Imtiaz Khan |
| sbMCB100       | Shafqat Ali |
+----------------+-------------+
8 rows in set (0.00 sec)
```

Task 4

Write a query to display the transaction details and customer name whose transaction are greater than 150000 in their saving account .

```
mysql> SELECT
    ->     t.transactionDetails, c.name
    -> FROM
    ->     transaction t
    -> INNER JOIN
    ->     savingaccounts s
    -> ON
    -> t.savingAccount_number = s.Account_number
    -> INNER JOIN
    ->     customer c
    -> ON
    -> s.customer_id = c.customer_id
    -> WHERE
    ->     t.transaction_amount > 150000;
+-----------------------------+--------+
| transactionDetails          | name   |
+-----------------------------+--------+
| credited amount of sold car | M. ALi |
+-----------------------------+--------+
1 row in set (0.00 sec)
```

Task 5

Display the name of the employees whose bank name start with ‘M’.

```
mysql> SELECT
    ->     e.name
    -> FROM
    ->     employee e
    -> INNER JOIN
    ->     branch b
    -> ON
    -> e.branch_id = b.branch_id
    -> INNER JOIN
    ->     bank bk
    -> ON
    -> b.bank_id = bk.bank_id
    -> WHERE
    ->     bk.bank_name LIKE 'M%';
+--------+
| name   |
+--------+
| Aslam  |
| Ahmad  |
| Zubair |
| Zaryab |
+--------+
4 rows in set (0.00 sec)
```

Task 6

Write a query to display loan account number of all the customers whose name starts with ‘I’ and end with ‘n’.

```
mysql> SELECT
    ->     l.Account_number
    -> FROM
    ->     loanaccounts l
    -> INNER JOIN
    ->     customer c
    -> ON
    -> l.customer_id = c.customer_id
    -> WHERE
    ->     c.name LIKE 'I%n';
+----------------+
| Account_number |
+----------------+
| lbUBL301       |
+----------------+
1 row in set (0.01 sec)
```

Task 7

Write a query to display employee name and transactions amount of all the employees. If an employee has not done any transaction then his transaction amount should be NULL.

```
mysql> SELECT e.name, t.transaction_amount
    -> FROM
    ->     employee e
    -> LEFT JOIN
    ->     transaction t
    -> ON
    -> e.eid = t.eid;
+---------+--------------------+
| name    | transaction_amount |
+---------+--------------------+
| Haider  |             -30000 |
| Ayesha  |             800000 |
| Anwar   |             900000 |
| Anwar   |             150000 |
| Kamran  |             -15500 |
| Kamran  |              12000 |
| Kamran  |              85000 |
| Kamran  |             175000 |
| Nouman  |              95000 |
| Tanveer |             100000 |
| Aslam   |               NULL |
| Ahmad   |               NULL |
| Zubair  |              27000 |
| Zubair  |              30000 |
| Zubair  |              20000 |
| Zaryab  |             -23000 |
| Zaryab  |             -32000 |
| Zaryab  |              15000 |
| Sara    |              30000 |
| Sara    |              70000 |
| Samreen |              28000 |
| Samreen |              60000 |
| Ali     |             -25000 |
| Ali     |              14000 |
| Ali     |              90000 |
| Ali     |              40000 |
| Ali     |              30000 |
| Sarmad  |             -26000 |
| Sarmad  |              20000 |
| Sarmad  |              95000 |
+---------+--------------------+
```

Task 8

Write a query to display customer name and account number of those customers who made transactions in year 2016.

```
mysql> SELECT DISTINCT
    ->     c.name, s.Account_number
    -> FROM
    ->     transaction t
    -> INNER JOIN
    ->     savingaccounts s
    -> ON
    -> t.savingAccount_number = s.Account_number
    -> INNER JOIN
    ->     customer c
    -> ON
    -> s.customer_id = c.customer_id
    -> WHERE
    ->     YEAR(t.transaction_date) = 2016;
+------------+----------------+
| name       | Account_number |
+------------+----------------+
| Imran Khan | sbUBL204       |
| M. Kareem  | sbUBL103       |
| Ali Ahmad  | sbABL107       |
| M. ALi     | sbABL303       |
+------------+----------------+
4 rows in set (0.00 sec)
```

Task 9

Write a query to display each employees bank branch details.

```
mysql> SELECT
    ->     e.name, b.branch_id, b.branch_address
    -> FROM
    ->     employee e
    -> INNER JOIN
    ->     branch b
    -> ON
    -> e.branch_id = b.branch_id;
+---------+-----------+-------------------------------+
| name    | branch_id | branch_address                |
+---------+-----------+-------------------------------+
| Haider  | ABL001    | Iqbal Town branch Lahore      |
| Ayesha  | ABL001    | Iqbal Town branch Lahore      |
| Anwar   | ABL002    | Johar Town branch Lahore      |
| Kamran  | ABL002    | Johar Town branch Lahore      |
| Nouman  | ABL003    | Barkat Market branch Lahore   |
| Tanveer | ABL003    | Barkat Market branch Lahore   |
| Aslam   | MCB001    | Moon Market branch Lahore     |
| Ahmad   | MCB001    | Moon Market branch Lahore     |
| Zubair  | MCB002    | 15, Jail road branch Lahore   |
| Zaryab  | MCB002    | 15, Jail road branch Lahore   |
| Sara    | UBL001    | Baghban pura branch Lahore    |
| Samreen | UBL001    | Baghban pura branch Lahore    |
| Ali     | UBL002    | Thokar niaz baig branch Lahore |
| Sarmad  | UBL002    | Thokar niaz baig branch Lahore |
+---------+-----------+-------------------------------+
14 rows in set (0.00 sec)
```

Task 10

Write a query to display all the names of employees with total amount of transaction till now.

```
mysql> SELECT
    ->     e.name, SUM(t.transaction_amount) AS total_transaction
    -> FROM
    ->     employee e
    -> LEFT JOIN
    ->     transaction t
    -> ON
    -> e.eid = t.eid
    -> GROUP BY
    -> e.eid;
+---------+-------------------+
| name    | total_transaction |
+---------+-------------------+
| Haider  |            -30000 |
| Ayesha  |            800000 |
| Anwar   |           1050000 |
| Kamran  |            256500 |
| Nouman  |             95000 |
| Tanveer |            100000 |
| Aslam   |              NULL |
| Ahmad   |              NULL |
| Zubair  |             77000 |
| Zaryab  |            -40000 |
| Sara    |            100000 |
| Samreen |             88000 |
| Ali     |            149000 |
| Sarmad  |             89000 |
+---------+-------------------+
14 rows in set (0.01 sec)
```